



Hashicorp Vault - secret management tool

Introduction

Well, If you're building a reliable application, the chances are that your functions require you to access secrets or any other types of sensitive information you're keeping.

These secrets may include:

- API keys
- Database credentials
- Encryption keys
- Sensitive configuration settings (email address, usernames, debug flags, etc.)
- Passwords

What is HashiCorp Vault and how does it work?

As many organizations migrate to the public cloud, a major concern has been how to best secure data, preventing it from unauthorized access or exfiltration.

Deploying a product like HashiCorp Vault gives you better control of your sensitive credentials and helps you meet cloud security standards.

In this post, we'll offer a high-level overview of what HashiCorp Vault is and share a few use cases — illustrating how it's an essential DevOps tool for your organization's cloud security.

What is HashiCorp Vault?

HashiCorp Vault is designed to help organizations manage access to secrets and transmit them safely within an organization. Secrets are defined as any form of sensitive credentials that need to be tightly controlled and monitored and can be used to unlock sensitive information. Secrets could be in the form of passwords, API keys, SSH keys, RSA tokens, or OTP.

HashiCorp Vault makes it very easy to control and manage access by providing you with a unilateral interface to manage every secret in your infrastructure. Not only that, you can also create detailed audit logs and keep track of who accessed what.

Vault

HashiCorp Vault is a tool for securely storing and accessing secrets. It provides a unified interface to secret while maintaining tight access control and logging a comprehensive audit log. It is a tool that secures user applications and base to limit the surface space and attack time in a breach. It gives an API that allows access to secrets based on policies. Any user of the API needs to verify and only see the secrets they are authorized to view.

Vault encrypts data using 256-bit AES with GCM.



How does HashiCorp Vault work?

HashiCorp Vault is a secrets management tool specifically designed to control access to sensitive credentials in a low-trust environment. It can be used to store sensitive values and at the same time dynamically generate access for specific services/applications on lease. Plus, Vault can be used to authenticate users (machines or humans) to make sure they're authorized to access a particular file.

Authentication can either be via passwords or using dynamic values to generate temporary tokens that allow you to access a particular path. Policies written using HashiCorp Configuration Language (HCL) are used to determine who gets what access.

Now that we've looked into what HashiCorp Vault is, let's review some of the things it's used for.

Vault

It can accumulate data in various backends such as Amazon DynamoDB, Consul, and much more. Vault supports logging to a local file for audit services, a Syslog server, or directly to a socket. Vault logs information about the client that acted, the clients IP address, the action, and at what time it was performed

Starting/restarting always involves one or more operators to unseal Vault. It works primarily with tokens. Each token is given to a policy that may constrain the actions and the paths. The key features of the Vault are:

- It encrypts and decrypts data without storing it.
- Vault can generate secrets on-demand for some operations, such as AWS or SQL databases.
- Allows replication across multiple data centers.
- Vault has built-in protection for secret revocation.
- Serves as a secret repository with access control details.

What is HashiCorp Vault used for?

HashiCorp Vault can be used in different ways; we've highlighted some of the main use cases below.

Secret management

HashiCorp Vault can be used to store any type of secrets, including sensitive environment variables, database credentials, API keys, and more, giving users control over who has access and who does not. Using Vault allows you to take full control of any sensitive credentials with the ability to rotate and revoke access at any time.

With HashiCorp Vault, you can rest assured that your credentials are secure, compared to storing plaintext files in your configuration management (for example).

Instead of storing plaintext files for all the world to see, you can have your application query vault read or the HashiCorp API, which protects the plaintext versions of those files.

Secrets are also easy to rotate and revoke; if an employee leaves your organization, you can easily and securely revoke their access.

What is HashiCorp Vault used for?

Identity-based access

HashiCorp Vault uses identity-based access to broker access to systems and secrets. When it comes to authenticating via identity, there are two major actors: humans and machines.

Managing access for humans is done through role-based access control (RBAC), granting permission and restricting access to either create and manage secrets or manage other users' access based on the secret value they are logged in with.

Managing access for machines on the other hand involves providing access to different servers or secrets. With the dynamic nature of HashiCorp Vault, you can create secrets that work temporarily and revoke access in the event of a breach. You can generate secrets on-demand for a particular system like Sensu, AWS, or Consul and generate a key pair with valid permission. After usage, the dynamic secrets generated will be automatically revoked.

What is HashiCorp Vault used for?

Data encryption

Vault provides “encryption as a service,” encrypting data in transit (with TLS) and at rest (using AES 256-bit CBC encryption). This protects sensitive data from unauthorized access in two major ways: as it travels across your network as well as in storage in your cloud and datacenters.

With centralized key management, it’s straightforward to update and roll out new keys across distributed infrastructure.



Installation

Install Vault

Vault must first be installed on your machine. Vault is available as source code, as a pre-compiled binary, or in packaged formats. To install Vault, find the appropriate package for your system and download it. Vault is packaged as a zip archive. After downloading Vault, unzip the package. Vault runs as a single binary named vault. Any other files in the package can be safely removed and Vault will still function.

The final step is to make sure that the vault binary is available on the PATH.



Verifying the Installation

After installing Vault, verify the installation worked by opening a new terminal session and checking that the vault binary is available. By executing vault, you should see help output similar to the following:

If you get an error that the binary could not be found, then your PATH environment variable was not setup properly. Please go back and ensure that your PATH variable contains the directory where Vault was installed.

Otherwise, Vault is installed and ready to go!

```
$ vault
```

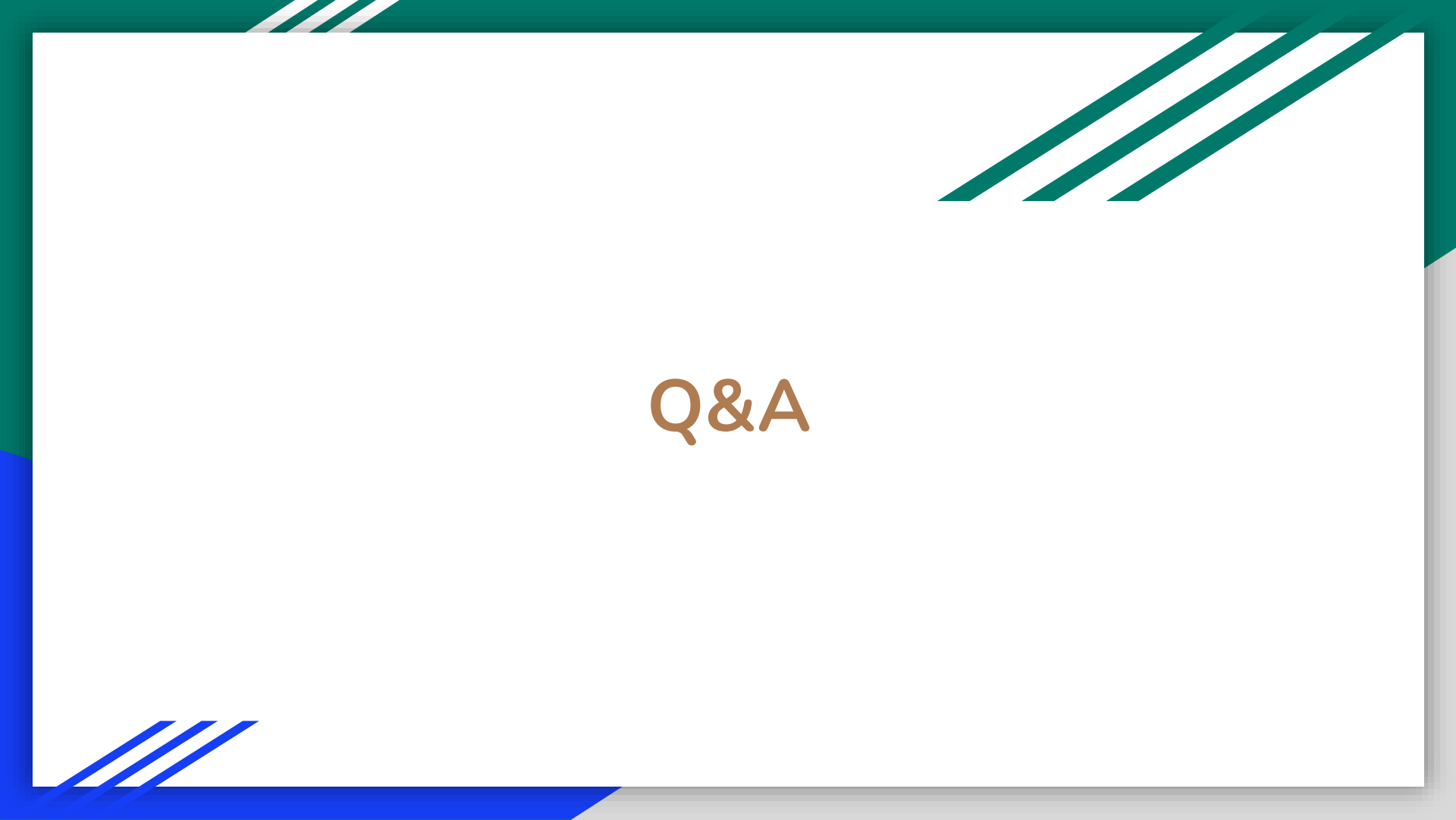
```
Usage: vault <command> [args]
```

Common commands:

read	Read data and retrieves secrets
write	Write data, configuration, and secrets
delete	Delete secrets and configuration
list	List data or secrets
login	Authenticate locally
server	Start a Vault server
status	Print seal and HA status
unwrap	Unwrap a wrapped secret

Other commands:

audit	Interact with audit devices
auth	Interact with auth methods
lease	Interact with leases
operator	Perform operator-specific tasks
path-help	Retrieve API help for paths
policy	Interact with policies
secrets	Interact with secrets engines
ssh	Initiate an SSH session
token	Interact with tokens



Q&A